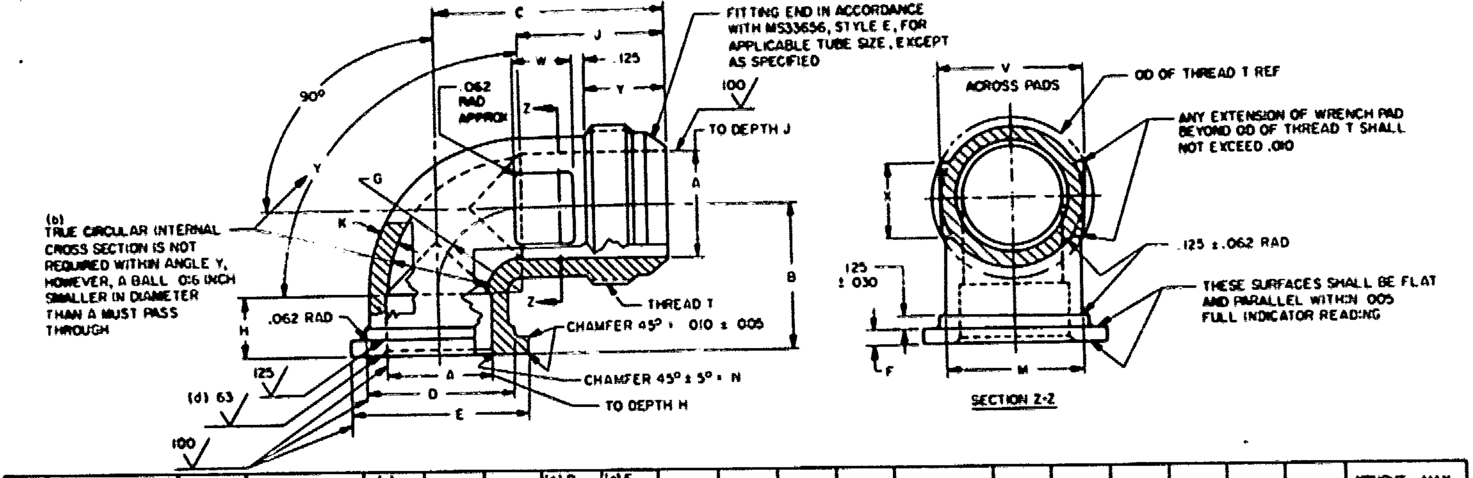
This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document

Review activities: NAVY - AS
USAF - 82

User activities:


C
J
W
.125
Y
.062
RAD
APPROX
Z
90°
Y
G
K
FITTING END IN ACCORDANCE WITH MS33656, STYLE E, FOR APPLICABLE TUBE SIZE, EXCEPT AS SPECIFIED
100
TO DEPTH J
A
B
(b) TRUE CIRCULAR INTERNAL CROSS SECTION IS NOT REQUIRED WITHIN ANGLE Y, HOWEVER, A BALL .06 INCH SMALLER IN DIAMETER THAN A MUST PASS THROUGH
H
.062 RAD
THREAD T
CHAMFER 45° × .010 ± .005
125
(d) 63
100
A
D
E
CHAMFER 45° ± 5° × N
TO DEPTH H
V
ACROSS PADS
OD OF THREAD T REF
ANY EXTENSION OF WRENCH PAD BEYOND OD OF THREAD T SHALL NOT EXCEED .010
X
.125 ± .062 RAD
.125 ± .030
THESE SURFACES SHALL BE FLAT AND PARALLEL WITHIN .005 FULL INDICATOR READING
F
M
SECTION Z-Z

DASH NO STEEL	DASH NO AL ALLOY	TUBING OD	THREAD T REF	(c) A ±.003 DIA	B +.047 -.000	C +.047 -.000	(c) D +.000 -.015 DIA	(c) E +.000 -.005 DIA	F +.000 -.005	G RAD REF	H +.047 -.000	J +.047 -.000	K MIN	K MAX	M ±.015 DIA	N DIA	V ±.015	W APPROX	X APPROX	Y ±.015	WEIGHT MAX AL ALLOY	WEIGHT MAX STEEL
-12	-012	.750	1-1/16-12UN-3A	.609	1 156	1 938	.938	1 250	.156	.594	.562	1 344	.125	.184	.922	.669	.891	.500	.562	.724	.12	.34
-12-16	-012-16					2.000		1 500				1 406									.6	.42
-16	-016	1.000	1-5/16-12UN-3A	.844	1 281	2 125	1 188			.719			.141	.176	1 172	.904	1 156	.562	.625	.771	.18	.50
-20	-020	1 250	1-5/8-12N-3A	1 078	1 500	2 375	1 500	1.844		.875	.625	1 900	.156	.221	1 484	1 130	1 438	.625	.750	.818	.32	.90
-20-24	-020-24				1 562	2 562		2 125	.188		.688	1 688									.36	1.00
-24	-024	1.500	1-7/8-12N-3A	1 312	1 688	2 688	1.750			1 000			.172	.234	1 734	1 372	1 710	.688		.943	.45	1 26
-32	-032	2 000	2-1/2-12UN-3A	1 781	2 062	3.375	2 375	2.750		1.312	.750	2.062	.203	.322	2.359	1.841	2 250	.750	.938	1.193	1.02	2.88
(a)-40	-040	2.500	3-12UN-3A	2 281	2 375	3.500	2 875	3.281		1.562	.812	1.938	.234	.333	2 859	2 341	2 812	.875	1.000	.969	1 37	3.84
(a)-48	-048	3.000	3-1/2-12UN-3A	2 781	2 625	3 875	3.375	3 781		1 812		2.062	.250	.344	3 359	2.841	3.344	1.000		1 047	1 82	5 10

Ⓐ ENTIRE STANDARD REVISED

(a) NOT TO BE USED UNLESS APPROVED BY PROCURING ACTIVITY
(b) INTERNAL FLOW PASSAGE AROUND BEND SHALL BE SMOOTH AND FREE FROM PROJECTIONS
(c) DIAMETERS A, D, AND E SHALL BE CONCENTRIC WITH EACH OTHER WITHIN .010 FULL INDICATOR READING.
(d) ANNULAR TOOL MARKS UP TO 63 RHR MAXIMUM WILL BE ALLOWED.

REDUCTION BY FORGING DRAFT ANGLE OF 7° MAXIMUM IS PERMISSIBLE.

POROSITY TEST. FINISHED CASTINGS SHALL NOT LEAK WHEN SUBJECTED TO 100-PSI AIR PRESSURE AND SUBMERGED IN WATER FOR 3 MINUTES MINIMUM.

FOR TUBE SIZES THRU 1 INCH. MAXIMUM OPERATING TEMPERATURE 275° F, MAXIMUM PRESSURE 1500 PSI.

FOR TUBE SIZES EXCEEDING 1 INCH: MAXIMUM OPERATING TEMPERATURE 275° F, MAXIMUM PRESSURE 500 PSI

MATERIAL: ALUMINUM-ALLOY FORGING, QQ-A-367 (2014), CLASS 5 OR 1. OPTIONAL MATERIAL: PERMANENT MOLD ALUMINUM ALLOY CASTING, QQ-A-596 OR ALUMINUM ALLOY BAR STOCK,
QQ-A-225/6 (2024)
STEEL FORGING, QQ-S-637 (1137) OR (1141).

UNLESS OTHERWISE SPECIFIED, SURFACE ROUGHNESS SHALL NOT EXCEED 100 MICROINCHES RHR. SEE USASI B 46.1

FINISH: ALUMINUM-ALLOY, ANODIZE, MIL-A-8625, TYPE 2, DYE LIGHT BLUE
STEEL, CADMIUM PLATE, QQ-P-416, TYPE 2, CLASS 3, DYE BLACK.

FITTINGS SHALL BE PERMANENTLY MARKED WITH MS PART NO. ON AN UNFINISHED SURFACE.

FOR NOMINAL USE ON FUEL AND OIL SYSTEMS.

BREAK ALL SHARP EDGES AND REMOVE ALL HANGING BURRS AND SLIVERS WHICH MIGHT BECOME DISLODGED UNDER USAGE.

DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED, TOLERANCES DECIMALS ±.005, ANGLES ±1/2°.

THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

FED. SUP CLASS
4730

APPROVED 9 Aug 57 REVISED Ⓐ 11 DEC 70

P.A. 82
Other Cust
Navy - AS
USAF - 82

TITLE
ELBOW, FLARED TUBE, FLANGED, SWIVEL, 90°

MILITARY STANDARD
MS20762

PROCUREMENT SPECIFICATION
NONE

SUPERSEDES: AN762

SHEET 1 OF 1

DD FORM 1 SEP 63 672-1 (coordination)

EDITION OF 1 SEP 57 MAY BE USED

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